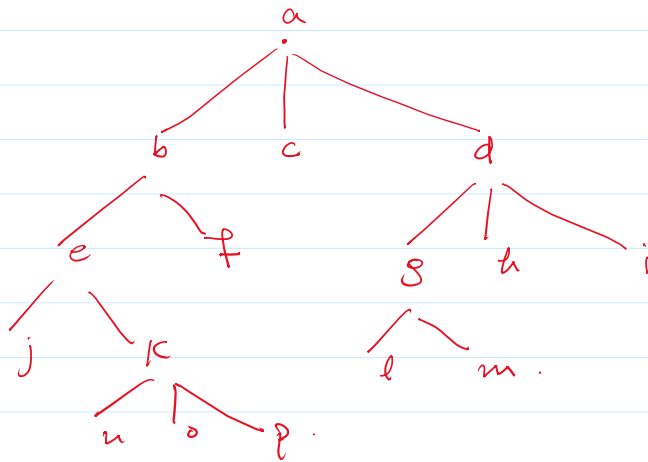
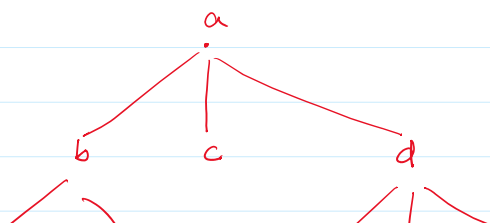
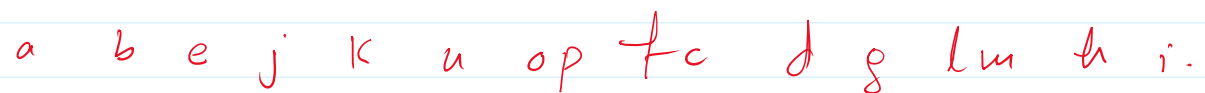
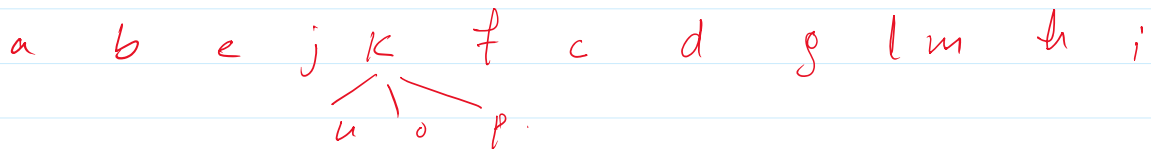
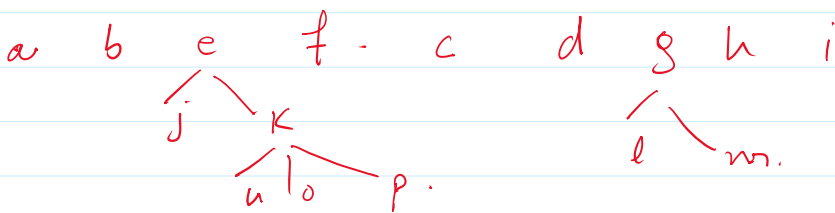
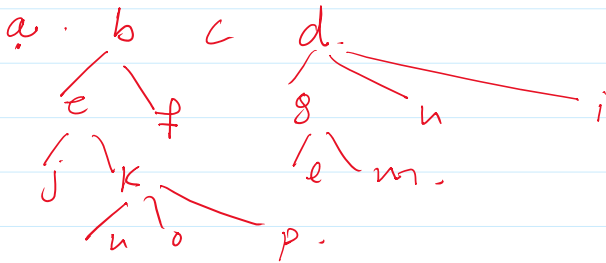


lecture 23.

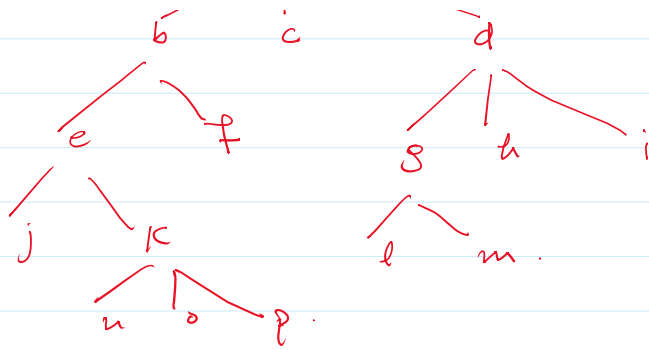
TREE TRAVERSAL.



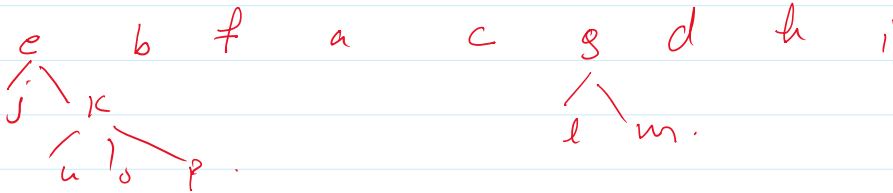
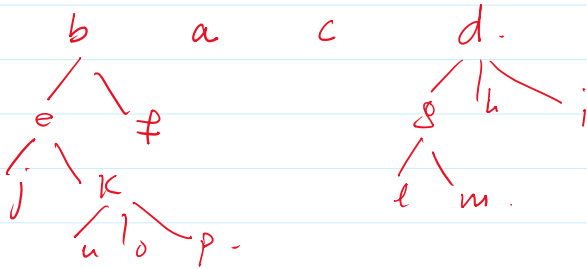
PRE ORDER.
N L R.



IN. ORDER.
L N R.



INORDER
LNR.



j e k b f a c l g m d h i

j e u k o p b f a c l g m d h i

Application of Tree Traversal.

- Expression Evaluation.

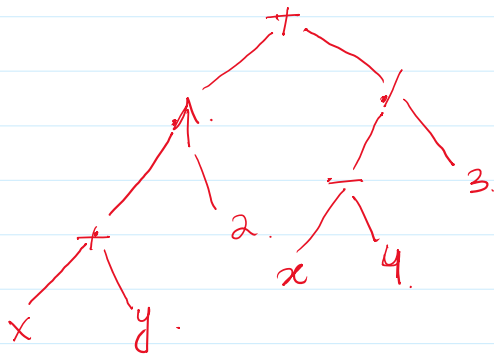
- Expression
p 657 :-
Ex 5

Tree :-

$((x+y)^2) + ((x-4)/3)$.



INORDER (INFIX NOTATION)



INORDER. (INFIX NOTATION)
L N R.

PREORDER. PREFIX NOTATION.

POSTORDER. POSTFIX NOTATION.

Prefix.

Ex 7
658.

+ - * 2 3 5 / ↑ 2 3 4.

+ - * 2 3 5 / 8 4

+ - * 2 3 5 2

+ - 6 5 2

+ 1 2 = 3.

213,

Ex 8
658.

Postfix.

7 2 3 * - 4. ↑ 9 3 / +.

7 6 - 4 ↑ 9 3 / +.

1 4 ↑ 9 3 / +.

1 9 3 / +.

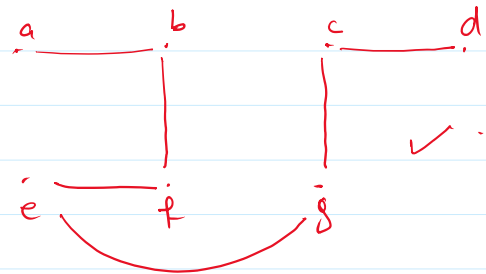
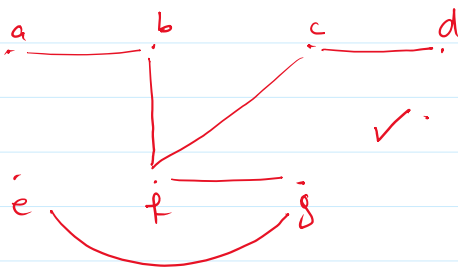
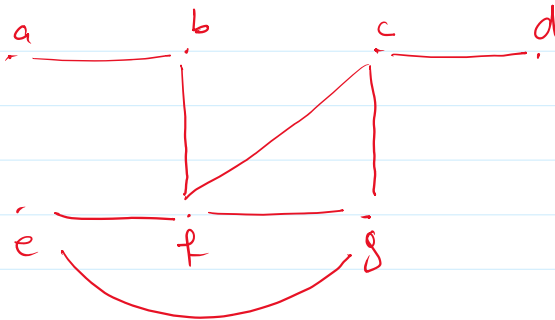
1 3 + = 4.

SPANNING TREE

- Connected
- Undirected.
- No Simple Circuit.
- Contains all Vertices.

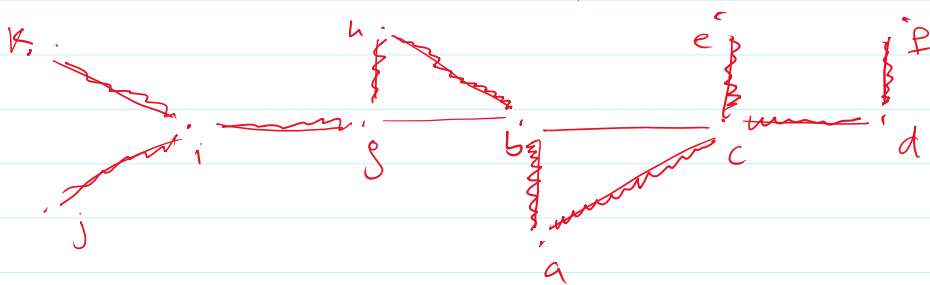
- No simple circuit.
- Contains all Vertices.
- Subgraph of Original Graph.

663.
Ex1

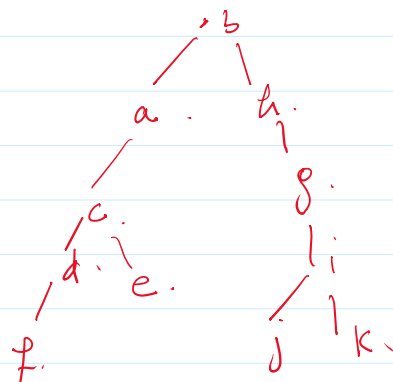


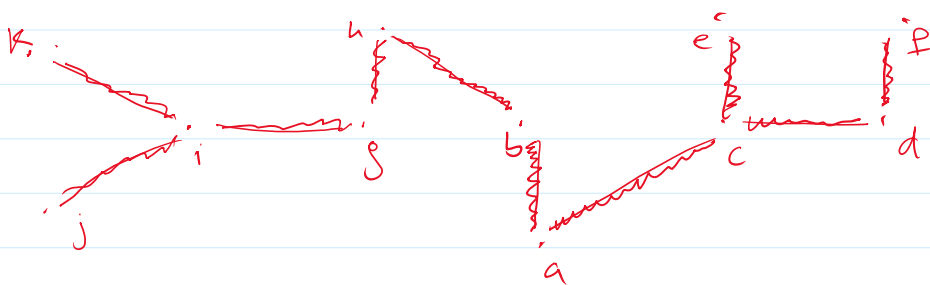
CONSTRUCTING SPANNING TREE.

- DEPTH FIRST.

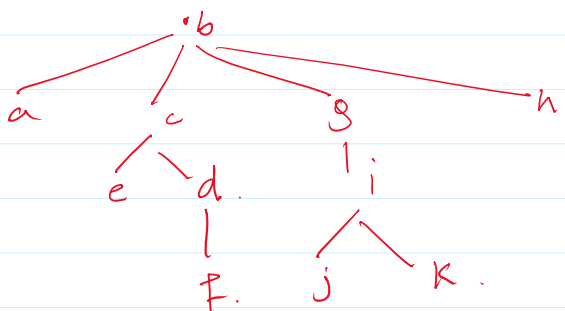
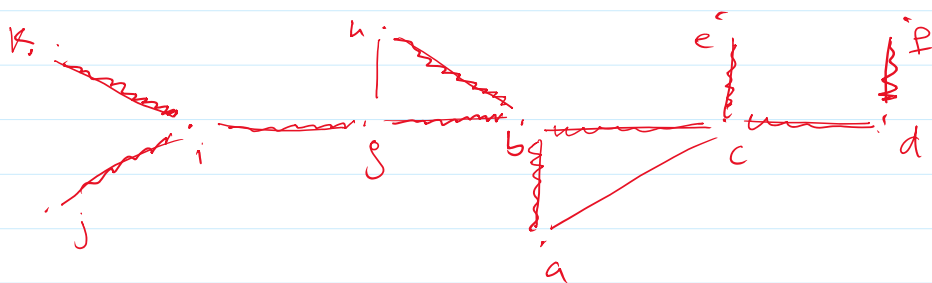


Dead END.
BackTrack.





- BREADTH FIRST.



Quiz 9

21-Nov-2025.

— Compute the following Prefix Notation.

$$\wedge \vee \wedge \rightarrow p q \vee p p.$$

$$p = F$$

$$q = T$$

$$\vee = F.$$

Quiz 10

21-Nov-2025.

— find spanning tree of C_8 & W_5 using depth first.

Solution Quiz 9.

$$\wedge \vee \wedge \rightarrow p q \vee p p.$$

$$p = F$$

$$q = T$$

$$\vee = F.$$

$$\wedge \vee \wedge \rightarrow F T F F F.$$

$$\wedge \vee \wedge T F F F$$

$$\wedge \vee F F F.$$

$$\wedge F F = F.$$

Quiz 10

21-Nov-2025.

— find spanning tree of C_8 & W_5 using depth first.

